

From Anomaly Detection to MobileNetV2: What I Learned Building a Wheat Rust Identification System

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Introduction

Wheat rust can affect crop health and agricultural productivity, making timely identification valuable for monitoring and decision support. During my MSc Artificial Intelligence research at Ulster University, I explored whether computer vision could distinguish healthy wheat leaves from leaves affected by brown rust or yellow rust.

The project began as an investigation into anomaly detection. The central question was attractive: if an autoencoder learned the appearance of healthy wheat leaves, could diseased leaves be detected as visual anomalies without requiring disease labels?

I later compared this approach with a supervised MobileNetV2 classifier. The two approaches produced very different results:

Approach	Task	Evaluation result
Convolutional autoencoder	Healthy versus disease anomaly	33.65% accuracy; 0.027 intended-direction ROC-AUC
MobileNetV2 classifier	Healthy, brown rust and yellow rust	98.78% accuracy; 0.988 weighted F1-score

This article explains what those results mean, why the supervised model performed much better and, most importantly, why a poor anomaly result must be investigated before concluding that the underlying method is unsuitable.

Dataset and evaluation setup

The experiment used the Wheat Rust Classification Dataset, containing three image classes: healthy leaves, brown rust and yellow rust.

The audited split contained:

Class	Training images	Held-out evaluation images
Brown rust	902	226
Healthy	1,116	279
Yellow rust	924	232
Total	2,942	737

The audit found no corrupt files or exact duplicate groups. It also identified 3,627 distinct image dimensions, showing substantial variation in the source images. Resizing was therefore necessary before batching images for model training.

The held-out split was used to compare both approaches. For the corrected anomaly experiment, the 1,116 healthy training images were separated into 892 images for model training, 112 for early-stopping validation and 112 for threshold calibration. The 737 held-out images were not used for model fitting or threshold selection. Because the evaluation images were still drawn from the same source dataset rather than independently collected in the field, the results should be interpreted as experimental performance—not evidence of real-world agricultural readiness.

Experiment 1: Learning “Healthy” With an Autoencoder

An autoencoder contains an encoder that compresses an input image into a smaller representation and a decoder that reconstructs the image. In reconstruction-based anomaly detection, the model is normally trained on examples of the normal class—in this case, healthy wheat leaves.

The intended assumption was:

1. healthy leaves should be reconstructed well;
2. diseased leaves should differ from the learned healthy distribution;
3. diseased leaves should therefore have higher reconstruction error;
4. a threshold on reconstruction error should separate healthy and anomalous images.

The corrected autoencoder experiment was trained for 15 epochs. Final training loss was 0.0061 and early-stopping validation loss was 0.0066. On the surface, this looked encouraging: the network had clearly learned to reconstruct images more accurately.

However, low reconstruction loss does not automatically mean strong anomaly separation.

Using a 95th-percentile reconstruction-error threshold of approximately 0.01105, calculated from the separate healthy calibration subset, the held-out evaluation produced:

- healthy recall: 88.89%;
- disease-anomaly recall: 0%;

- overall accuracy: 33.65%;
- intended-direction ROC-AUC: 0.027;
- inverted-direction diagnostic ROC-AUC: 0.973.

The model effectively labelled nearly everything as healthy. It did not identify the 458 diseased evaluation images as anomalies.

Why the anomaly pipeline produced a poor result

It would be tempting to conclude that “autoencoders do not work for wheat rust.” The evidence does not support such a broad statement. Several technical explanations must first be tested.

1. The reconstruction errors moved in the opposite direction from the assumption

The corrected evaluation confirmed that the labels and ROC-AUC calculation were oriented as intended: disease was the positive class and larger reconstruction error represented greater anomalousness. Nevertheless, diseased images consistently received lower reconstruction errors than healthy images.

The measured reconstruction-error distributions were:

- healthy mean: 0.00662; median: 0.00600;
- brown-rust mean: 0.00154; median: 0.00146;
- yellow-rust mean: 0.00114; median: 0.00106;
- combined disease mean: 0.00134; median: 0.00120.

The inverted-direction AUC of 0.973 shows that reconstruction error contained a strong separating signal, but in the opposite direction from the intended healthy-only anomaly model. Simply reversing the score after examining disease labels would turn the method into a supervised calibration choice. It would not validate the original assumption that previously unseen disease should produce greater reconstruction error.

This result may indicate that the disease images were visually easier to reconstruct—for example, because they contained more consistent colours, textures, backgrounds or image-acquisition characteristics—while the healthy class contained greater natural diversity. These are hypotheses that require visual and source-level investigation.

2. Reconstruction quality and anomaly separation are different objectives

The autoencoder was optimised to reduce reconstruction loss. It was not directly optimised to maximise the distance between healthy and diseased samples.

A sufficiently expressive autoencoder may reconstruct disease patterns reasonably well, even when it has only been trained on healthy examples. Previous anomaly-detection

research has identified this limitation: anomalies can sometimes receive low reconstruction error, weakening their usefulness as outliers.

3. Healthy wheat leaves contain substantial natural variation

Healthy images can vary in lighting, background, orientation, scale, colour and camera quality. If the healthy training distribution is broad, its reconstruction-error distribution may overlap with that of diseased leaves.

Rust can also appear as local texture and colour changes rather than a complete change to the image. A whole-image reconstruction score may be dominated by the larger areas of normal leaf and background, reducing sensitivity to small lesions.

4. A single global threshold may be inadequate

The corrected threshold was approximately 0.01105 and was calibrated without using the held-out evaluation set. However, it was an upper-tail threshold because the intended method assumed that disease would cause higher reconstruction error. Since every diseased evaluation image fell below that disease threshold, the detector missed the entire disease class.

A stronger analysis would report:

- histograms or density plots for each class;
- precision-recall and ROC curves;
- threshold sensitivity;
- confusion matrices at several candidate thresholds;
- confidence intervals obtained through resampling.

5. Global pixel error may ignore the most useful disease features

Mean reconstruction error treats every region of the image as part of one score. Wheat rust, however, is often represented by spatially local colour and texture patterns. Patch-level errors, feature-space distances or segmentation-guided scores may preserve these signals better than a single image-level pixel loss.

Experiment 2: MobileNetV2 transfer learning

The second approach treated the problem as supervised three-class classification. Instead of learning only the appearance of healthy leaves, the model learned explicitly from labelled examples of:

- brown rust;
- healthy wheat;
- yellow rust.

MobileNetV2 is a lightweight convolutional neural-network architecture based on inverted residual blocks and linear bottlenecks. Transfer learning allows features learned from a large

general image dataset to be adapted to a new task, reducing the need to learn every visual representation from the beginning.

The classifier was trained for 10 epochs. Training accuracy increased from 84.50% to 98.98%, while held-out accuracy reached 98.78%. The final validation loss was 0.0410.

Class-level results

Brown rust

Precision: 98.65% | Recall: 97.35% | F1-score: 98.00% | Support: 226

Healthy

Precision: 98.93% | Recall: 99.64% | F1-score: 99.29% | Support: 279

Yellow rust

Precision: 98.71% | Recall: 99.14% | F1-score: 98.92% | Support: 232

The model correctly classified 728 of the 737 held-out images. Its nine errors were:

- three brown-rust images predicted as healthy;
- three brown-rust images predicted as yellow rust;
- one healthy image predicted as brown rust;
- two yellow-rust images predicted as brown rust.

Why MobileNetV2 performed better

Direct supervision

The most important difference is that MobileNetV2 received the information needed for the task. It was trained to minimise classification error across all three labelled classes. The autoencoder, by contrast, was trained to reconstruct healthy images and relied on the indirect assumption that disease would create higher error.

Transferable visual features

The pretrained network begins with useful representations of edges, shapes, colours and textures. Fine-tuning or adapting these features can help distinguish rust patterns without learning visual primitives from scratch.

Class-specific decision boundaries

The classifier learned separate boundaries for healthy, brown-rust and yellow-rust images. This is particularly valuable when both diseases remain visually similar to ordinary wheat leaves but differ through local patterns.

Better alignment between training objective and deployment question

The application asks: “Which of these three known classes best describes this image?” Supervised classification directly matches that question. Reconstruction-based anomaly detection answers a different one: “How unusual is this image compared with the learned healthy distribution?”

Neither question is universally better. They serve different deployment conditions.

What I would improve next

Investigate the reversed reconstruction behaviour

The corrected experiment verified score orientation, label encoding and leakage-free threshold calculation. The next priority is to plot class-level reconstruction distributions, inspect low- and high-error images and investigate whether background, image dimensions, compression, colour statistics or collection sources explain the reversed behaviour. The experiment should then be repeated on an independently sourced test set.

Compare stronger anomaly baselines

Future experiments could include:

- anomaly scoring in pretrained feature space;
- patch-level reconstruction errors;
- one-class support vector machines;
- isolation forests on deep image embeddings;
- memory-augmented or constrained autoencoders;
- synthetic anomaly generation for threshold calibration.

Strengthen the supervised evaluation

The high MobileNetV2 result is promising, but it needs more rigorous testing. I would add:

- a completely independent test split;
- stratified cross-validation where appropriate;
- testing on images from different farms, devices and lighting conditions;
- checks for near-duplicates and source leakage, not only exact duplicates;
- model calibration and uncertainty analysis;
- Grad-CAM or another interpretability technique to verify that predictions rely on leaf symptoms rather than backgrounds.

Treat the application as a research demonstration

The deployed Streamlit interface shows how the model behaves on uploaded images, but it is not a replacement for expert agricultural diagnosis. Field validation, broader data collection and careful monitoring would be required before any operational use.

Key lessons

This project changed how I interpret machine-learning results.

First, a falling training loss proves only that the model is improving against its training objective. It does not prove that the objective is aligned with the real task.

Second, a surprising metric should be investigated before being turned into a conclusion. After removing evaluation leakage, the anomaly detector produced an intended-direction AUC of 0.027 and an inverted diagnostic AUC of 0.973. This revealed strong separation in the wrong direction and raised an important dataset-shortcut question that accuracy alone would have hidden.

Third, supervised transfer learning performed strongly because labelled disease examples, pretrained visual features and a classification objective all aligned with the target task.

Finally, reproducible reporting should include unsuccessful experiments. Explaining what failed, what remains uncertain and what should be tested next is more useful than presenting one successful accuracy figure without context.

Project links

- [Live Wheat Rust AI demonstration](#)
- [GitHub repository](#)
- [MobileNetV2 research paper](#)
- [TensorFlow transfer-learning guide](#)
- [Research discussing limitations of reconstruction-based anomaly detection](#)

This article documents an independent academic and technical experiment. The reported results were generated on the project's held-out evaluation split. They have not yet been independently peer reviewed or validated in operational agricultural settings.